

SOHAM SANTOSH KALGUTKAR

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Education

Fr. Conceicao Rodrigues College of Engineering <i>B.E. in Computer Engineering, CGPA: 9.15 (up to Semester 6)</i>	2023 – Expected 2027 <i>Mumbai</i>
Dadar Pioneer Junior College <i>Higher Secondary School Certificate, 82%, MHT-CET: 96.767 Percentile</i>	2021 – 2023 <i>Mumbai</i>
IES Manik Vidyamandir <i>Secondary School Education (ICSE), 96%</i>	2021 <i>Mumbai</i>

Skills & Technologies

Languages: C, C++, Python, JavaScript, SQL, Java

Frameworks & Libraries: React.js, Node.js, FastAPI, Express.js, Streamlit, Boto3, Flex, Bison

Databases & Cloud: MongoDB, DynamoDB, AWS (Lambda, S3, SNS), Git/GitHub

Projects

Course Allocation System <i>React.js, Python, FastAPI, MongoDB, JWT</i>	Github
– Architected a full-stack web application for students and faculty using FastAPI and React, featuring a dual-role JWT authentication system that secures student elective submissions and grants teachers administrative control over the allocation process. – Engineered a robust MongoDB data model managing student preferences across 7 academic categories, with server-side validation to enforce data integrity and prevent duplicate course selections during the submission workflow. – Automated academic resource planning by developing a backend reporting module that generates Excel/CSV enrollment summaries, enabling faculty to perform real-time demand analysis and streamline final course allocation.	
Cloud-Sentry: Serverless Malware Scanning Pipeline (POC) <i>AWS, Python (Boto3), VirusTotal API</i>	Github
– Architected a serverless, event-driven security pipeline using AWS Lambda and S3 to automate real-time malware detection; developed a decoupled backend that intercepts file uploads and routes them to isolated clean or quarantine storage zones. – Implemented a privacy-focused scanning logic by integrating the VirusTotal API via SHA-256 file hashing, keeping file content within the secure cloud environment while performing lookups against 70+ antivirus engines. – Enforced cloud security best practices via IAM Least Privilege policies, a DynamoDB-backed audit trail, and SNS integration for immediate administrative alerts upon threat detection.	
jqlite: Full-Stack JSON Query Engine <i>C (Flex/Bison), Node.js</i>	Private Repository : IP Protected
– Engineered a high-performance JSON query engine in C using Flex/Bison, implementing a custom <i>Filter Promotion</i> AST optimization that achieved a 2× runtime improvement and 500× faster field lookups via $O(1)$ hash-table integration. – Developed an end-to-end web-based visualization system with Node.js and Monaco Editor, maintaining query latency below 200ms and ensuring 100% functional correctness through a rigorous 135-case automated test suite.	

Awards & Patents

1st Place: Data Analytics Hackathon, IES MCRC College

Patent Published: Project K.A (Kitchen Automate): An IoT & AI-based Smart Kitchen Trolley